

Experiment 6

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Class: SE3

Config.php

```
<?php
declare(strict_types=1);

session_start([
    'cookie_httponly' => true,
    'cookie_samesite' => 'Lax',
    'use_strict_mode' => true,
]);

const DB_HOST = '127.0.0.1';
const DB_NAME = 'home_crud';
const DB_USER = 'root';
const DB_PASS = '';

const REMEMBER_COOKIE_EXPIRES = 2592000; // 30 days

function db(): PDO
{
    static $pdo = null;

    if ($pdo instanceof PDO) {
        return $pdo;
    }

    $dsn = sprintf('mysql:host=%s;dbname=%s;charset=utf8mb4', DB_HOST, DB_NAME);

    $pdo = new PDO($dsn, DB_USER, DB_PASS, [
        PDO::ATTR_ERRMODE => PDO::ERRMODE_EXCEPTION,
        PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC,
        PDO::ATTR_EMULATE_PREPARES => false,
    ]);
}
```

```

        return $pdo;
    }

function e(?string $value): string
{
    return htmlspecialchars((string) $value, ENT_QUOTES, 'UTF-8');
}

function redirect(string $path): never

```

```

{
    header('Location: ' . $path);
    exit;
}

function set_flash(string $type, string $message): void
{
    $_SESSION['flash'] = [
        'type' => $type,
        'message' => $message,
    ];
}

function get_flash(): ?array
{
    $flash = $_SESSION['flash'] ?? null;
    unset($_SESSION['flash']);

    return $flash;
}

function csrf_token(): string
{
    if (empty($_SESSION['csrf_token'])) {
        $_SESSION['csrf_token'] = bin2hex(random_bytes(32));
    }

    return $_SESSION['csrf_token'];
}

```

```
function verify_csrf(): void
{
    $token = $_POST['csrf_token'] ?? '';

    if (!is_string($token) || !hash_equals($_SESSION['csrf_token'] ?? '', $token)) {
        http_response_code(419);
        exit('Invalid CSRF token.');
```

```
function normalize_input(array $source): array
{
```

```
    return [
        'name' => trim((string) ($source['name'] ?? '')),
        'email' => trim((string) ($source['email'] ?? '')),
        'phone' => trim((string) ($source['phone'] ?? '')),
        'city' => trim((string) ($source['city'] ?? '')),
    ];
}
```

```
function validate_user(array $input): array
{
    $errors = [];

    if ($input['name'] === '') {
        $errors['name'] = 'Name is required.';
    } elseif (mb_strlen($input['name']) > 100) {
        $errors['name'] = 'Name must be 100 characters or fewer.';
    }

    if ($input['email'] === '') {
        $errors['email'] = 'Email is required.';
    } elseif (!filter_var($input['email'], FILTER_VALIDATE_EMAIL)) {
        $errors['email'] = 'Please enter a valid email address.';
    } elseif (mb_strlen($input['email']) > 150) {
        $errors['email'] = 'Email must be 150 characters or fewer.';
    }
}
```

```

    if ($input['phone'] !== '' && !preg_match('/^[0-9+\-\s]{7,20}$/', $input['phone']))
    {
        $errors['phone'] = 'Phone should contain only numbers, spaces, + or - signs.';
    }

    if ($input['city'] !== '' && mb_strlen($input['city']) > 100) {
        $errors['city'] = 'City must be 100 characters or fewer.';
    }

    return $errors;
}

function remember_user_input(array $input): void
{
    $expires = time() + REMEMBER_COOKIE_EXPIRES;

```

```

    setcookie('remember_name', $input['name'], [
        'expires' => $expires,
        'path' => '/',
        'httponly' => true,
        'samesite' => 'Lax',
    ]);

    setcookie('remember_email', $input['email'], [
        'expires' => $expires,
        'path' => '/',
        'httponly' => true,
        'samesite' => 'Lax',
    ]);
}

function set_last_visit_cookie(): void
{
    setcookie('last_visit', date('Y-m-d H:i:s'), [
        'expires' => time() + REMEMBER_COOKIE_EXPIRES,
        'path' => '/',
        'httponly' => true,
        'samesite' => 'Lax',
    ]);
}

```

```

    ]);
}

function previous_input(): array
{
    return $_SESSION['old_input'] ?? [];
}

function previous_errors(): array
{
    return $_SESSION['errors'] ?? [];
}

function flash_form_state(array $input, array $errors): void
{
    $_SESSION['old_input'] = $input;
    $_SESSION['errors'] = $errors;
}

function clear_form_state(): void
{
    unset($_SESSION['old_input'], $_SESSION['errors']);
}

```

Create.php

```

<?php
declare(strict_types=1);

require __DIR__ . '/config.php';

if ($_SERVER['REQUEST_METHOD'] !== 'POST') {
    redirect('/Home/project/index.php');
}

verify_csrf();

$input = normalize_input($_POST);
$errors = validate_user($input);

```

```

if ($errors) {
    flash_form_state($input, $errors);
    set_flash('error', 'Please fix the highlighted fields and try again.');
```

`redirect('/Home/project/index.php');
}

try {
 $statement = db()->prepare(
 'INSERT INTO users (name, email, phone, city) VALUES (:name, :email, :phone,
:city)'
);
 $statement->execute($input);

 remember_user_input($input);
 set_flash('success', 'User created successfully.');`

```

} catch (PDOException $exception) {
    flash_form_state($input, []);
    set_flash('error', 'Unable to save the user. The email may already exist.');
```

`}

redirect('/Home/project/index.php');`

Delect.php

```

<?php
declare(strict_types=1);

require __DIR__ . '/config.php';

if ($_SERVER['REQUEST_METHOD'] !== 'POST') {
    redirect('/Home/project/index.php');
}

verify_csrf();

$id = filter_input(INPUT_POST, 'id', FILTER_VALIDATE_INT);

if (!$id) {
```

```

        set_flash('error', 'Invalid user selected for deletion.');
```

```

        redirect('/Home/project/index.php');
```

```

    }

    try {
        $statement = db()->prepare('DELETE FROM users WHERE id = :id');
        $statement->execute(['id' => $id]);
        set_flash('success', 'User deleted successfully.');
```

```

    } catch (PDOException $exception) {
        set_flash('error', 'Unable to delete the user right now.');
```

```

    }

    redirect('/Home/project/index.php');
```

Index.php

```

<?php
declare(strict_types=1);

require __DIR__ . '/config.php';

$_SESSION['dashboard_visits'] = (int) ($_SESSION['dashboard_visits'] ?? 0) + 1;
```

```

$flash = get_flash();
$formData = previous_input();
$errors = previous_errors();
clear_form_state();

$editId = filter_input(INPUT_GET, 'edit', FILTER_VALIDATE_INT);
$editUser = null;
$dbError = null;
$users = [];

try {
    $pdo = db();

    if ($editId) {
        $statement = $pdo->prepare('SELECT id, name, email, phone, city, created_at,
```

```

updated_at FROM users WHERE id = :id');
    $statement->execute(['id' => $editId]);
    $editUser = $statement->fetch();

    if (!$editUser) {
        set_flash('error', 'The selected user does not exist.');
```

```

        redirect('/Home/project/index.php');
    }
}

    $users = $pdo->query('SELECT id, name, email, phone, city, created_at, updated_at
FROM users ORDER BY id DESC')->fetchAll();
} catch (PDOException $exception) {
    $dbError = 'Database connection failed. Create the database from schema.sql and
check config.php credentials.';
}

$isEditing = is_array($editUser);
$prefill = [
    'name' => $formData['name'] ?? ($editUser['name'] ?? ($_COOKIE['remember_name'] ??
    '')),
    'email' => $formData['email'] ?? ($editUser['email'] ?? ($_COOKIE['remember_email']
    ?? '')),
    'phone' => $formData['phone'] ?? ($editUser['phone'] ?? ''),
    'city' => $formData['city'] ?? ($editUser['city'] ?? ''),
];

```

```

$previousVisit = $_COOKIE['last_visit'] ?? 'This is your first visit.';
set_last_visit_cookie();
?>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>PHP MySQL User Manager</title>
    <link rel="stylesheet" href="/Home/project/styles.css">
</head>

```



```

<body>
<main class="page">
  <section class="hero">
    <div>
      <p class="eyebrow">PHP, MySQL, Sessions, Cookies</p>
      <h1>User Management Dashboard</h1>
      <p class="hero-copy">
        Create, review, update, and delete users with server-side validation,
        prepared statements, session flash messages, CSRF protection, and
        cookie-based form memory.
      </p>
    </div>
    <div class="hero-stats">
      <div class="stat-card">
        <span>Session visits</span>
        <strong><?= (int) $_SESSION['dashboard_visits'] ?></strong>
      </div>
      <div class="stat-card">
        <span>Last visit</span>
        <strong><?= e($previousVisit) ?></strong>
      </div>
    </div>
  </section>

  <?php if ($flash): ?>
    <div class="alert alert-<?= e($flash['type']) ?>"><?= e($flash['message'])
?></div>
  <?php endif; ?>

  <?php if ($dbError): ?>
    <div class="alert alert-error"><?= e($dbError) ?></div>

```

```

<?php endif; ?>

<section class="grid">
  <article class="panel">
    <div class="panel-header">
      <h2><?= $isEditing ? 'Update User' : 'Add User' ?></h2>
      <?php if ($isEditing): ?>
        <a class="link-button" href="/Home/project/index.php">Cancel

```

Edit

```
        <?php endif; ?>
    </div>

    <form method="post" action="<?=$isEditing ? '/Home/project/update.php' :
'/Home/project/create.php' ?>" novalidate>
        <input type="hidden" name="csrf_token" value="<?=$csrf_token() ?>">
        <?php if ($isEditing): ?>
            <input type="hidden" name="id" value="<?=(int) $editUser['id']
?>">

            <?php endif; ?>

            <label for="name">Full Name</label>
            <input id="name" name="name" type="text" maxlength="100" value="<?=$prefill['name'] ?>" required>
            <?php if (isset($errors['name'])): ?>
                <small class="error-text"><?=$errors['name'] ?></small>
            <?php endif; ?>

            <label for="email">Email Address</label>
            <input id="email" name="email" type="email" maxlength="150" value="<?=$prefill['email'] ?>" required>
            <?php if (isset($errors['email'])): ?>
                <small class="error-text"><?=$errors['email'] ?></small>
            <?php endif; ?>

            <label for="phone">Phone Number</label>
            <input id="phone" name="phone" type="text" maxlength="20" value="<?=$prefill['phone'] ?>" placeholder="+91 9876543210">
            <?php if (isset($errors['phone'])): ?>
                <small class="error-text"><?=$errors['phone'] ?></small>
            <?php endif; ?>

            <label for="city">City</label>
```

```
        <input id="city" name="city" type="text" maxlength="100" value="<?=$prefill['city'] ?>" placeholder="Ahmedabad">
        <?php if (isset($errors['city'])): ?>
            <small class="error-text"><?=$errors['city'] ?></small>
        <?php endif; ?>
```

```
<button class="primary-button" type="submit"><?= $isEditing ? 'Update
User' : 'Save User' ?></button>

</form>

</article>

<article class="panel">
  <div class="panel-header">
    <h2>Stored Users</h2>
    <span class="badge"><?= count($users) ?> records</span>
  </div>

  <div class="table-wrap">
    <table>
      <thead>
        <tr>
          <th>ID</th>
          <th>Name</th>
          <th>Email</th>
          <th>Phone</th>
          <th>City</th>
          <th>Created</th>
          <th>Actions</th>
        </tr>
      </thead>
      <tbody>
        <?php if (!dbError && $users): ?>
          <?php foreach ($users as $user): ?>
            <tr>
              <td><?= (int) $user['id'] ?></td>
              <td><?= e($user['name']) ?></td>
              <td><?= e($user['email']) ?></td>
              <td><?= e($user['phone']) ?></td>
              <td><?= e($user['city']) ?></td>
              <td><?= e((string) $user['created_at']) ?></td>
              <td class="actions">
                <a class="link-button"
href="/Home/project/index.php?edit=<?= (int) $user['id'] ?>">Edit</a>
                <form method="post"

```

```

action="/Home/project/delete.php" onsubmit="return confirm('Delete this user?');"
        <input type="hidden" name="csrf_token"
value="<?= e(csrf_token()) ?>">
        <input type="hidden" name="id" value="<?= (int)
$user['id'] ?>">
        <button class="danger-button"
type="submit">Delete</button>
    </form>
</td>
</tr>
<?php endforeach; ?>
<?php elseif ($dbError): ?>
    <tr>
        <td colspan="7" class="empty-state">The user list is
unavailable until the database connection is fixed.</td>
    </tr>
<?php else: ?>
    <tr>
        <td colspan="7" class="empty-state">No users found yet. Add
one using the form.</td>
    </tr>
<?php endif; ?>
</tbody>
</table>
</div>
</article>
</section>
</main>
</body>
</html>

```

SQL

```

CREATE DATABASE IF NOT EXISTS home_crud CHARACTER SET utf8mb4 COLLATE
utf8mb4_unicode_ci;
USE home_crud;

CREATE TABLE IF NOT EXISTS users (
    id INT UNSIGNED AUTO_INCREMENT PRIMARY KEY,

```

```
name VARCHAR(100) NOT NULL,  
email VARCHAR(150) NOT NULL UNIQUE,  
phone VARCHAR(20) DEFAULT NULL,  
city VARCHAR(100) DEFAULT NULL,  
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP  
);
```

Style.css

```
:root {  
  --bg: #f5efe6;  
  --panel: rgba(255, 255, 255, 0.92);  
  --panel-border: rgba(37, 33, 31, 0.08);  
  --text: #1f1a17;  
  --muted: #6d5c52;  
  --primary: #b85042;  
  --primary-dark: #8e3d31;  
  --danger: #b42318;  
  --success: #166534;  
  --error: #991b1b;  
  --shadow: 0 24px 60px rgba(76, 53, 40, 0.14);  
}  
  
* {  
  box-sizing: border-box;  
}  
  
body {  
  margin: 0;  
  min-height: 100vh;  
  font-family: "Trebuchet MS", "Segoe UI", sans-serif;  
  color: var(--text);  
  background:  
    radial-gradient(circle at top left, rgba(184, 80, 66, 0.18), transparent 30%),  
    radial-gradient(circle at bottom right, rgba(31, 26, 23, 0.15), transparent  
25%),  
    linear-gradient(135deg, #f8f3ee, #eadfce 55%, #d9c5af);  
}
```

```
.page {
```

```
  width: min(1180px, calc(100% - 32px));
```

```
  margin: 0 auto;
```

```
  padding: 32px 0 48px;
```

```
}
```

```
.hero {
```

```
  display: flex;
```

```
  justify-content: space-between;
```

```
  gap: 24px;
```

```
  align-items: flex-start;
```

```
  margin-bottom: 24px;
```

```
}
```

```
.eyebrow {
```

```
  margin: 0 0 8px;
```

```
  color: var(--primary);
```

```
  font-size: 0.9rem;
```

```
  font-weight: 700;
```

```
  letter-spacing: 0.08em;
```

```
  text-transform: uppercase;
```

```
}
```

```
h1,
```

```
h2 {
```

```
  margin: 0;
```

```
}
```

```
.hero-copy {
```

```
  max-width: 680px;
```

```
  color: var(--muted);
```

```
  line-height: 1.6;
```

```
}
```

```
.hero-stats {
```

```
  display: grid;
```

```
  gap: 12px;
```

```
    min-width: 240px;
}

.stat-card,
.panel {
    background: var(--panel);
```

```
    border: 1px solid var(--panel-border);
    border-radius: 22px;
    box-shadow: var(--shadow);
    backdrop-filter: blur(8px);
}
```

```
.stat-card {
    padding: 18px;
}
```

```
.stat-card span {
    display: block;
    color: var(--muted);
    font-size: 0.88rem;
    margin-bottom: 8px;
}
```

```
.stat-card strong {
    display: block;
    font-size: 1rem;
    line-height: 1.4;
}
```

```
.grid {
    display: grid;
    grid-template-columns: 360px 1fr;
    gap: 24px;
}
```

```
.panel {
    padding: 24px;
}
```

```
.panel-header {
  display: flex;
  justify-content: space-between;
  align-items: center;
  gap: 12px;
  margin-bottom: 20px;
}

form {
```

```
  display: grid;
  gap: 12px;
}

label {
  font-weight: 700;
  font-size: 0.95rem;
}

input {
  width: 100%;
  border: 1px solid rgba(31, 26, 23, 0.12);
  border-radius: 14px;
  padding: 12px 14px;
  font-size: 0.98rem;
  transition: border-color 0.2s ease, box-shadow 0.2s ease;
}

input:focus {
  outline: none;
  border-color: rgba(184, 80, 66, 0.8);
  box-shadow: 0 0 0 4px rgba(184, 80, 66, 0.12);
}

.primary-button,
.danger-button,
.link-button {
  border: 0;
  border-radius: 999px;
  padding: 11px 16px;
```



```
font-weight: 700;
cursor: pointer;
text-decoration: none;
text-align: center;
transition: transform 0.15s ease, opacity 0.15s ease, background 0.15s ease;
}

.primary-button:hover,
.danger-button:hover,
.link-button:hover {
  transform: translateY(-1px);
}
```

```
.primary-button {
  background: var(--primary);
  color: #fff;
}

.primary-button:hover {
  background: var(--primary-dark);
}

.danger-button {
  background: #fee2e2;
  color: var(--danger);
}

.link-button {
  background: rgba(184, 80, 66, 0.12);
  color: var(--primary-dark);
}

.alert {
  margin-bottom: 18px;
  padding: 14px 16px;
  border-radius: 16px;
  font-weight: 700;
}
```

```
.alert-success {
  background: rgba(22, 101, 52, 0.12);
  color: var(--success);
}

.alert-error {
  background: rgba(153, 27, 27, 0.1);
  color: var(--error);
}

.error-text {
  color: var(--error);
  margin-top: -4px;
}
```

```
.badge {
  display: inline-flex;
  align-items: center;
  justify-content: center;
  min-width: 96px;
  padding: 6px 12px;
  border-radius: 999px;
  background: rgba(31, 26, 23, 0.08);
  color: var(--muted);
  font-size: 0.88rem;
  font-weight: 700;
}
```

```
.table-wrap {
  overflow-x: auto;
}
```

```
table {
  width: 100%;
  border-collapse: collapse;
}
```

th,

```
td {
  text-align: left;
  padding: 14px 10px;
  border-bottom: 1px solid rgba(31, 26, 23, 0.08);
  vertical-align: top;
}

th {
  color: var(--muted);
  font-size: 0.88rem;
  text-transform: uppercase;
  letter-spacing: 0.04em;
}

.actions {
  display: flex;
  gap: 10px;
  align-items: center;
  flex-wrap: wrap;
}

.actions form {
  display: inline-flex;
}

.empty-state {
  text-align: center;
  color: var(--muted);
  padding: 36px 12px;
}

@media (max-width: 920px) {
  .hero,
  .grid {
    grid-template-columns: 1fr;
    display: grid;
  }
}
```

```
@media (max-width: 640px) {  
  .page {  
    width: min(100% - 20px, 100%);  
    padding-top: 20px;  
  }  
  
  .panel,  
  .stat-card {  
    border-radius: 18px;  
    padding: 18px;  
  }  
  
  th,  
  td {  
    padding-inline: 6px;  
  }  
}
```

```
<?php
```

```
declare(strict_types=1);
```

Update.php

```
require __DIR__ . '/config.php';

if ($_SERVER['REQUEST_METHOD'] !== 'POST') {
    redirect('/Home/project/index.php');
}

verify_csrf();

$id = filter_input(INPUT_POST, 'id', FILTER_VALIDATE_INT);
$input = normalize_input($_POST);
$errors = validate_user($input);

if (!$id) {
    set_flash('error', 'Invalid user selected for update.');
```

```
    redirect('/Home/project/index.php');
}

if ($errors) {
    flash_form_state($input, $errors);
    set_flash('error', 'Please fix the highlighted fields before updating.');
```

```
    redirect('/Home/project/index.php?edit=' . $id);
}

try {
    $statement = db()->prepare(
        'UPDATE users
        SET name = :name, email = :email, phone = :phone, city = :city
        WHERE id = :id'
    );
    $statement->execute($input + ['id' => $id]);

    remember_user_input($input);
    set_flash('success', 'User updated successfully.');
```

```
} catch (PDOException $exception) {
    flash_form_state($input, []);
}
```

```
set_flash('error', 'Unable to update the user. The email may already exist.');
```

```
redirect('/Home/project/index.php?edit=' . $id);
```

```
}
```

```
redirect('/Home/project/index.php');
```

PHP, MYSQL, SESSIONS, COOKIES

User Management Dashboard

Create, review, update, and delete users with server-side validation, prepared statements, session flash messages, CSRF protection, and cookie-based form memory.

Session visits
4

Last visit
2026-04-06 12:02:37

Add User

Full Name

RAHUL

Email Address

maguvof0jed2@maximail.vip

Phone Number

+91 9876543210

City

Ahmedabad

Save User

Stored Users

1 records

ID	NAME	EMAIL	PHONE	CITY	CREATED	ACTIONS
1	RAHUL	maguvof0jed2@maximail.vip	+91 87349 73923		2026-04-06 15:32:37	Edit Delete